Dissertation work map and current model chapter outline

Updated: 26 October 2026. Originally prepared after PD44 on 30 September; now reconciled with PD47, PD48, PD49, the post-brief discussion, and the saved manuscript and NCAA work.
Purpose: Keep the dissertation structure, manuscript status, advisor direction, completed investigations, and next scientific tasks in one place.
Status: Working plan for Charles. Alex's documented proposals, Charles's subsequent decisions, and our proposed next steps are distinguished below. This is not an approved final table of contents.

The dissertation connects the science of success with network science: people perform and pursue advancement within groups, and the composition of those groups may affect who receives scarce opportunities. Army, basketball, education, and academic careers supply possible evidence and examples. The dissertation should remain organized around the scientific question rather than become an account of one domain.

The model chapter has a substantially revised introduction and development of its components, but the later analytical and empirical argument remains unfinished. Alex's latest request is to show what the model explains: why each ingredient is needed, when a downturn appears, and how its location and curvature change with the parameters. Introducing the ingredients and deriving local sensitivities are foundations for that task, not its completion.

Our current side investigation is the BPM basketball comparison documented in Entry 7 of the NCAA Evidence Walkthrough. It has not been inserted into the manuscript. The distinction matters: the working chapter has newer prose, the walkthrough has newer empirical work, and this map connects their status. The separate Levine–Gates (LG) group-formation chapter remains a possibility, not an agreed addition.

Read this map as the current planning record. The filename retains PD44 so existing links work. The transition memo and October 18 LOO and HERO methods handoff remain historical sources. They do not override later decisions or the saved Word manuscript. The memo now contains an additional handoff dated October 25 that acknowledges the live Word document is more current than the October 20 clean copy. Its older clean-copy instruction and section/notation descriptions should not override the current manuscript or our later editing decisions.

1. What Alex proposed

PD44 on 30 September

Alex described three central **content chapters**, with introductory and surrounding material bringing the dissertation to roughly five chapters. His "first," "second," and "third" refer to those content chapters, not settled dissertation chapter numbers.

1. **Establish the empirical phenomenon.** Explain the relationship among individual performance, group environment, and advancement. At 02:52 he recommends bringing together evidence from several datasets, including comparisons among people with similar performance and the band of excellence.
2. **Develop the model.** Explain its construction, assumptions, parameter behavior, and completed investigations in dissertation-level detail. At 05:56–08:43 he distinguishes this detailed treatment from the much shorter account an eventual paper would contain.
3. **Connect the model to the evidence.** At 06:31 he identifies maximum likelihood estimation and fitting the model to data as the next content chapter. This is a research destination, not a declaration that the fitting bridge is already complete.

At approximately 09:10–11:31, Alex also supports preserving the Army work on defining success, promotion timing, and survival or competing risks. He suggests that this could become a chapter before the peer-effects discussion. He does not settle its final placement.

At 12:24 he recommends starting the writing with the model chapter. Writing should reveal specific missing analyses; it should not wait for every empirical problem to be resolved.

Primary source: PD44 transcript.

PD45 on 7 October and the email on the first draft

Alex's later guidance changes the model chapter's exposition, not the distinction between scoring and selection. PD45 recommends starting with the simplest performance-based selection model and adding complexity as the reader encounters a reason for it. Assignment comes later as a way to construct groups for simulations; the advancement model can begin with groups already observed.

His email likewise says not to introduce ASSIGN as an equal part of the scientific model, to reduce repetition, and to move the internal development-history account out of the main scientific narrative. Detailed mathematics still belongs in the dissertation when it explains the model or supports a claim.

Charles clarified that Alex's concern about "separating environmental effects from later selection" should not be read as merging SCORE and SELECT into a single operation. They remain mathematically distinct; explaining advancement requires considering their combination.

Sources: PD45 transcript; Alex's email reproduced by Charles in this conversation; the October 8 change memo. The email is identified as conversational evidence rather than an independently located email file.

Framing recorded by Charles

Dissertation Tone records Charles's account of a conversation with Alex. It emphasizes the bridge between network science and the science of success, the balance of readable narrative with mathematical development, and the need to return from domain examples to the general phenomenon. This is Charles's written account, not a verbatim transcript.

PD47 on 16 October

Alex describes the draft as a substantial foundation but finds its prose too mechanical. At 00:00–03:17, his criterion is that the first sentences of the paragraphs should tell a coherent story on their own. Remove contrasts that only make sense in the history of our conversations. Introduce the situation and the scientific question in the order a new reader encounters them. At 04:13–05:20, he recommends a whole-section flow pass after the paragraph-level content has been settled, using the actual latest text rather than an agent's remembered version.

At 22:01–24:07, Alex distinguishes unfinished science from unfinished writing. The missing scientific contribution includes fitting back to data and, more importantly, predictions that can be tested. His assessment is not that editing alone will complete the paper.

The timeline discussion at 08:00–18:30 considers a December defense, a January alternative, and possible arrangements for more writing time. These were options to investigate, not an approved extension, confirmed defense date, or verified statement of university requirements. Track administrative decisions separately when Charles confirms them.

Source: PD47 transcript.

PD48 on 20 October

Alex explicitly places a possible survival-analysis chapter **later**, after a simpler empirical measure has introduced the phenomenon (00:01–00:37). This supersedes the earlier working suggestion in this map that the detailed Army methods chapter might precede the empirical peer-effects chapter.

At 00:43–03:08, Charles proposes predicting where the HERO pattern should appear, including which own-performance band is relevant under different scarcity conditions. Alex strongly supports this as a prospective contribution while explicitly declining to guarantee that the prediction will succeed. At 03:08–04:48, he also proposes predictions involving system noise and the relationship between observed performance and advancement. These are research goals, not verified cross-domain findings.

Source: PD48 transcript.

Feedback after the lab briefing

Alex wants the defense presentation to spend much less time introducing the machinery and much more time on why it is needed and what it teaches. At 00:01–01:48, he suggests reducing the model introduction to roughly a quarter of its briefing length. At each stage, compare the simpler model with the relevant data or target behavior, identify what it misses, and explain why the next ingredient addresses that deficiency. His request is for an evidentiary argument, not an assumption that the audience will infer necessity. At 02:06–03:50, he supports the player graphics and improved narrative while asking for less text and less elementary explanation.

The second part concerns Army reproducibility. At 05:02–09:25, he prioritizes an authorized analysis-ready export when access returns: a table with manufactured person and unit identifiers, score, time, and necessary covariates. Discuss the permitted release with the responsible Army contacts. Hashing is a proposed approach to discuss, not proof of anonymity or authorization to release data. Screenshots alone do not supply the reproducible analysis he wants. No contact, export, or access restoration is claimed here.

Source: post-brief feedback transcript. The filename is dated October 22; the transcription header says October 23. We retain that discrepancy rather than silently resolving it.

PD49 on 23 October

Alex asks us to define the curve the model is meant to explain and why that curve matters scientifically (00:05–03:19). His proposed summaries are the position of its local maximum and the coefficient of its squared term, describing downward curvature. He then asks how model parameters move those quantities. The curve, axis, conditioning, and scale must be specified before those summaries are meaningful.

At 03:19–05:32, he asks for a demonstration of what simpler models cannot produce, preferably analytically. Fitting a quadratic to binned empirical rates is explicitly distinguished from deriving a response from the model. The existing local probability comparisons are useful but do not yet carry the argument through the full selection mechanism.

At 06:05–10:20 and 11:58–12:40, he asks us to substitute the dependence of congestion on own performance into the model and work toward selection-probability curves in own performance and in the chosen group-environment measure. His expectation that higher performance should not lower selection prospects is a conjecture or desired model property to examine, not an established theorem. The current manuscript already identifies parameter regimes where the focal score derivative can be negative; selection probabilities require further analysis of all affected scores and the normalizing denominator.

At 10:28–11:55, Alex challenges the use of different environmental measures across domains. The objective is a common, defensible scientific target and comparable congestion specification, not to select whichever axis yields a downturn in each dataset. Full-group mean, LOO mean, and roster pressure must remain distinct while that choice is unresolved.

At 12:40–15:32, he asks for parameter-dependent curve families and quantitative changes in the peak and curvature. Directly compared curves should share axes. Larger mosaics can remain working diagnostics, but they do not replace a focused comparison in the chapter. At 15:40–16:24, Charles and Alex agree that the old multi-panel figures under discussion are not the intended final presentation. Their removal or replacement remains a manuscript task; agreement to remove is not evidence that removal has occurred.

Source: PD49 transcript.

2. Our current proposed dissertation arrangement

The following is **our updated planning arrangement for discussion**, not a chapter list approved by Alex. We retain the working label “Chapter 4” for the manuscript already in progress. We should choose final numbers only after deciding whether the Army methods and LG contributions stand alone.

Introduction and literature

Set out the scientific question and connect work on success, peer environments, group composition, and scarce advancement. Define what the dissertation contributes beyond demonstrating that strong individuals tend to be surrounded by other strong individuals.

Basis: Alex explicitly calls for introduction and literature material in PD44. The wording and scope here are our proposed organization.

Measuring advancement and the empirical phenomenon

Establish how success is measured and then describe the observed patterns across suitable domains. Explain population definitions, time windows, group membership, performance measures, and which comparisons hold own performance approximately fixed. Describe similarities and differences across domains without assuming all show one identical effect.

Basis: This is Alex’s first central content block. Our proposed treatment keeps measurement and source limitations close to the claims they qualify.

Army measurement at this stage: Introduce a clearly labeled, interpretable empirical outcome measure and its limits. Reserve the fuller treatment of promotion timing, attrition, censoring, survival, and competing risks for a later methodological chapter or substantial later section. PD48 explicitly supports the later placement; the final chapter count and the estimator actually used remain to be verified.

The congestion and advancement model

This is the current **Chapter 4**, described in Section 3 below. Begin with the naïve performance-only benchmark, introduce uncertain selection, develop congestion, examine its mathematics, and then explain how artificial groups are generated.

Basis: Alex's model content block, revised through PD47–49 and Charles's subsequent editing decisions. Assignment remains a simulation device within this chapter even if LG receives a separate chapter elsewhere.

Possible chapter on LG and group formation

Study how metadata-based attachment preference and prescribed roster capacities produce group composition. Then compare clearly identified alternatives that change the opportunity rule or relax fixed roster sizes.

Basis: Charles's later proposal and the separate LG exploration specification. We have not verified an explicit endorsement from Alex of a separate LG chapter. The executive summary calls it a possibility for discussion.

Proposed placement: After the core advancement model and before the model-to-data chapter is a useful working option. It would let the main model chapter explain the assignment algorithm briefly, while the LG chapter develops group formation as its own scientific subject. Placement remains open; avoid duplicating the same derivation and figures in both chapters.

Connecting the model to evidence

Explain the historical fitting work, what it estimates, and what is still needed to connect those estimates to the intended model. Distinguish empirical fitting, simulation-based parameter recovery, and counterfactual claims.

Basis: Alex's third central content block. A completed exact-K refit or validated counterfactual analysis is not currently established by the sources reviewed for this map. Chapter 4's present Section 4.8 introduces this bridge; it need not duplicate an eventual full fitting chapter.

Later development of advancement measurement

Return to the simplified empirical outcome and explain what a fuller treatment of time, censoring, promotion and departure changes. This is the later survival-analysis contribution discussed in PD48. It should improve the scientific interpretation rather than repeat the early empirical chapter. Identify the actual historical estimator before attributing existing Army curves or bars to Fine–Gray regression.

Synthesis and conclusions

Bring the theoretical and empirical contributions together, state their limits, and identify the next scientific questions. A synthesis chapter is our proposed completion of the structure, not a separately verified detailed instruction from Alex in PD44.

This arrangement deliberately leaves the final chapter count open. Adding both Army methods and LG as standalone chapters would expand Alex's initial rough five-chapter conception; we should make that an explicit discussion rather than silently relabel his original proposal.

3. What is actually in the current manuscript

The authoritative editing copy is Model Chapter Shared Working Draft, inspected on October 26. This review reads current text with tracked deletions excluded and includes Alex's retained comments; it does not accept changes, resolve comments, or modify the Word file. The Markdown, LaTeX, clean saves and older backups are reference versions. Charles often makes different choices from a proposed paragraph, so always reread the saved working copy.

Editing agreement: Normally supply precise, paste-ready text one paragraph at a time, stating exactly what it replaces. Charles edits the working document, keeps comments and Track Changes, and periodically requests a backup before accepting revisions himself. Direct Word editing requires his request. Maintain Verdana 10 pt body, the agreed heading and caption styles, and editable Cambria Math equations and symbols at 11 pt when making such edits. Supply bare LaTeX symbols for copying without added delimiter parentheses. Audit whole-section flow after the paragraph pass.

4.1 Individual performance and success in groups

Present: A natural-language account of strong peers helping development while intensifying competition for recognition and opportunities; the general scientific question; a simple-to-complex roadmap. This reflects the earlier narrative revisions.

Still to reconcile: The roadmap currently says fitting comes before assignment, while the actual sections put assignment in 4.6 and fitting in 4.8. Choose and synchronize the intended order in a later editing pass. More importantly, sharpen the empirical target and why it matters, as requested in PD49, without asserting that the same downturn has already been established in every domain.

4.2 Selecting the highest performing individuals

Present: Performance and the eligible population, scarcity, separate SCORE and SELECT operations, and the deterministic top-K benchmark. The two current subsections are "Performance and the selection population" and "The score and the selection rule."

Still needed: Tie this benchmark to the precisely defined empirical/model target. Demonstrating a monotone own-performance rule is not, by itself, a proof about every aggregated team-environment curve. State the population, grouping, conditioning and selection assumptions for any claimed impossibility result.

4.3 Allowing uncertainty in selection

Present: "Gibbs-weighted selection without replacement" and "Selection temperature and limiting behavior." The text introduces weighting before temperature, describes the current eligible pool including the focal candidates, distinguishes next-draw probability from final inclusion among K winners, and identifies the Plackett–Luce form. The probability ratio, limits and score-scale invariances are developed. The opening asks how much a score advantage changes the chance of selection.

Status: The substantial collaborative prose pass is present. This is not a claim that every comment has been resolved or that the section is final. Its existing distinction between draw probability and final inclusion must survive the PD49 analytical development.

4.4 Introducing local competitive congestion

Present: The order is full-group mean T_j, a concise LOO peer-mean definition, then competitive relevance and shared roster pressure. Developmental benefits are acknowledged but outside this fixed-performance model. The text distinguishes descriptive peer measures from congestion used in scoring. Roster pressure is introduced as a third congestion measure; averaging relevance is explained as competitive concentration. The longer LOO material is in Appendix 4A.5. The local-quota cancellation argument is in Appendix 4A.8.

Where the paragraph-level pass paused: Around 4.4.3, after the competitive-concentration explanation, at the paragraph beginning “This definition makes a specific modeling choice: congestion is a shared feature of the group environment.” That wording is still present. Our unresolved issue is that sharedness belongs to this roster-pressure specification, not to every possible congestion measure. The later four-measure notation discussion was tabled for the presentation; it is not an adopted renaming scheme.

Still needed: State the intended full model in the currently empty “Summary” placeholder, respond to Alex’s request for evidence about the limitations of T_j, and reconcile the earlier three-measure exposition with PD49’s request for one coherent cross-domain target. Do not quietly replace C_LOO,i, rename C_j, or declare a winning measure before that decision.

4.5 Mathematical behavior and parameter regimes

Present: Sigmoid sensitivities and limits; the maximum relevance sensitivity at the threshold; score crossings; the own-performance score derivative and its parameter bound; scarcity and selection margins; and the distinction between the teammate changing pressure and the player near the selection boundary. Appendix 4A.7 carries additional sensitivity-through-selection development. The section also distinguishes changed winners from a downturn and warns that broad ability bands do not fix own performance exactly.

Still needed: The PD49 development from these results to defined selection curves, including how the peak and curvature depend on parameters. The existing derivative of the score and the pairwise next-draw probability ratio do not finish that task. Alex’s comments explicitly request consequences through the full model, clearer lambda tradeoffs, and a possible derivation relating theta to K/N. Such a relationship is not yet established merely by choosing an empirical quantile.

Figures remain historical: The text still describes a top-7% PPM comparison in Figure 4-4 and 2011–2021 PPM player-season bands in Figure 4-5, with their original binning and sample definitions. The latest BPM experiments have not replaced them. Keep those populations separate from the final-season BPM cohort until we choose the manuscript figures deliberately.

4.6 Generating artificial groups with controlled sorting

Present: Assignment is introduced as a device for controlled simulations, with fixed capacities, sequential centroid updates, and the H_sort variance decomposition. The random-assignment derivation is retained in Appendix 4A.6. Observed rosters do not require an assignment fit to use the advancement mechanism.

Still needed: Carry the revised narrative style through the section and repair remaining draft fragments during the normal editing pass. Keep the brief model-chapter treatment separate from the possible LG chapter. The lab presentation’s graphic success does not settle the scientific necessity of a particular sorting mechanism.

4.7 What the completed mechanism experiments establish

Present: Historical deterministic simulations and figures concerning winner replacement, penalty strength, binning, assignment preference and scarcity, with limitations stated. The manuscript still contains the multi-row Figures 4-7 and 4-8. These paragraphs are not evidence that a new, comprehensive parameter exploration has been completed.

Still needed: Decide which verified historical comparisons answer the new questions and replace or remove the rejected multi-panel presentation. Put directly compared curve families on shared axes. State evaluation criteria before interpreting simulations. Alex’s comments question the repeated “completed” language; review which claims the exact saved implementation and outputs actually support. Do not discard underlying results merely because their current presentation is unsuitable.

4.8 The bridge to empirical fitting

Present: A candid account of the historical season-wise softmax/Bernoulli fit, its player-season eventual-draft labels, normalization, threshold preprocessing, and difference from sequential exact-K selection. Saved coefficients are described as belonging to the historical specification; no completed exact-K refit is claimed.

Still needed: A clear likelihood for the scientific outcome actually being fitted, with observation units, timing, conditioning, capacity and uncertainty specified. Alex asks for the fitting specifics rather than repeated motivation or a second description of the model with renamed parameters. Retain consequential qualifications, put implementation history where it belongs, and do not treat historical coefficients as automatically transferable.

Appendices and document maintenance

Appendix 4A retains implementation qualifications, fitting history, alternative congestion algebra, random sorting, sensitivity through selection, and local-quota cancellation. Appendix 4B is the source register. Charles asked to postpone a full appendix organization pass; that remains open. Preserve scholarly citations as well as repository provenance.

This review also sees out-of-order cached equation labels in 4.4.1. They need a separate field/numbering check when Charles requests it; no equations, captions, comments, numbering or formatting were changed here. Existing comments may remain even where the text addresses them, so neither the comment count nor the section’s existence proves completion.

4. The LG chapter possibility and its recorded status

The account below retains the October 13 planning assessment. The decision log still supports the documented distinction between composition experiments and later scoring/selection work. This October 26 update is not a fresh audit of all LG outputs or a new decision to add a chapter.

Scientific purpose

The original LG model combines attachment preference based on individual and group metadata with **fixed prescribed roster sizes**. Remaining seats represent available assignment opportunities, not attraction to high degree. A separate chapter could establish what this combination contributes to sorting, variation within and between teams, and the concentration of credible competitors.

Relaxing fixed rosters changes the model. The exploration should name each alternative and explain its kernel instead of treating every variant as LG.

What is documented now

The October 6 executive summary and handoff set out the proposed study. The newer LG decision log records subsequent implementation and Charles's approvals. It supersedes the handoff's statements that the working code has yet to be built.

A working composition notebook, Python modules, and settings files now exist. The decision log reports completed saved divisions for the first fixed-roster composition comparison. It also records contender-count diagnostics, an uncapped extension, alternative initial attachment signals and mean-versus-sum signals, and an optional one-plus-members opportunity multiplier. That last option introduces a size-dependent preference and must be distinguished from original fixed-roster LG and from uncapped metadata-only assignment.

The earlier planning assessment inspected the implementation inventory and decision record; this update reread the decision log but did not rerun experiments or independently audit those saved results. The existence of code or a plotting option is not evidence that a scientific conclusion has been established.

Proposed chapter development

1. Define fixed-roster LG and its scientific question.
2. Explain and compare the assignment opportunity rules while holding populations and relevant constraints constant.
3. Present verified composition and sorting results with repetitions and uncertainty described correctly.
4. Treat relaxed capacities and other attachment kernels as explicit extensions, identifying which assumption changes in each comparison.
5. Connect selected group-formation findings to congestion and advancement only where the scoring and selection analyses have actually been completed.
6. Use synthetic parameter recovery to diagnose fitting when the generating and fitted probability models are specified and aligned.

The last two items remain planned stages in the reviewed materials, not completed results certified here. Matching-model recovery should precede claims about fitting data generated under another selection rule. This map does not authorize new sweeps, fits, or high-performance-computing jobs.

Decision still needed

Charles and Alex should decide whether the verified LG findings support a distinct chapter with its own question and conclusion. Until then, reserve the possibility and preserve the work, while maintaining the short assignment explanation in Chapter 4.

5. Empirical work alongside the manuscript

The NCAA sequence we have actually followed

The NCAA Evidence Walkthrough, updated October 25, is our printable experiment journal. It has seven entries and 13 pages. It records the original performance-band overlay, the own-performance curve, the highest-bin subdivision, the low-minutes investigation, narrower BPM bands, peer-environment comparisons, and the aligned-threshold experiment. Its earlier entries remain a record of those earlier settings, not definitions of every later run.

The PPM shared versus LOO roster-pressure notebook preserves the original panel 7/8/9 cohorts. The BPM companion introduced final-season BPM bands, initially top 3% and next 3%, later top 2% and next 2%. That companion ranks the focal bands by raw BPM while calculating pressure from season-standardized BPM. Its latest saved baseline is `bpm_runs/20261025_184918_282428`, with theta quantile 0.98 and gamma 6.

Current separate experiment: BPM aligned theta and performance bands. At Charles's request, this starts with BPM alone. Both theta and the band ranks use season-standardized BPM among the same 19,289 final-season candidates. Pressure is still calculated on full available eligible team-season rosters before selecting the focal bands. The three bands are top 2%, top 10% excluding top 2%, and top 30% excluding top 10%.

The saved run `20261025_191833_184458` uses gamma 6, theta quantile 0.98, 16 quantile or equal-width bins, rank-first ties, and no clipping or tail compression. Theta is 2.0315. Exactly 386 candidates exceed it and none equal it. Counts are:

- Top 2%: 386 players, 191 ever drafted, 49.5%.
- Next 8%: 1,543 players, 250 ever drafted, 16.2%.
- Next 20%: 3,858 players, 65 ever drafted, 1.7%.

The high-pressure Jackal region has a higher draft rate than the middle Squid region under both pressure definitions in all three bands at these settings. There are local fluctuations, but this comparison does not supply the broad high-pressure downturn we anticipated. No new significance tests were run. The change to standardized ranking replaces 23 members of the previous 386-person top band; the new theta reference and wider lower bands change too. This is not a metric-only or single-parameter comparison.

Limits that remain visible: BPM is available for 2011–2021 in these sources, not the full PPM 2009–2021 window. K = 510 counts ever-drafted final-season candidates; K/N = 2.64% is not an observed annual quota. The top, middle and lower bands retain 75, 101 and 246 players with fewer than 100 recorded season minutes. Missing teammate BPM is omitted. A broad band does not hold own performance exactly fixed, and a fixed top percentile is not automatically a set of draft locks or a domain-independent band of excellence. Theta is chosen here, not fitted or set equal to a validated annual capacity.

Quantile bins are formed separately within each band and axis, so the same bin number does not mean the same numerical environment. Equal-width pressure edges are shared. Ties can be split by the agreed rank-first rule. A tie-preserving comparison remains parked for later; it has not silently replaced the requested binning.

For exact results use a named run and its `provenance.json`, not a notebook title or a “latest” pointer alone. The saved PPM settings cell currently says theta quantile 0.97 while its latest-run pointer records 0.98; this map notes the mismatch without rerunning or modifying that separate branch. The original October 23 defaults in the old map are historical, not current settings.

Evidence: aligned-run provenance, cohort counts, regional counts, and Entry 7 of the walkthrough. No empirical findings in this subsection were inserted into the manuscript by this update.

Army and the cross-domain comparison

The October 18 NVector to SVector methods handoff remains the detailed record of the CODA reconciliation. In the inspected exports, `pool_minus_mean` is own rating minus the peer mean, not the peer mean itself. CODA reports newer direct-LOO plots; the exact outputs and a matching refit still require identification. The inspected time curves accumulate events within bins and the HERO bars summarize their endpoints. Those outputs alone do not establish a Fine–Gray regression.

Charles recalls Fine–Gray work and expected access to the authoritative code and data. That historical-method question remains open until the generating cells, inputs, estimator and output can be matched. Do not infer that access has returned simply because a week has elapsed. The post-brief request for a reproducible, authorized analysis table is a priority when it does. Compare Army, basketball and tenure only after stating each axis, outcome, denominator, time horizon and treatment of unresolved outcomes. Similar shapes do not establish equivalent probabilities or one validated mechanism.

Romania

The October 25 selected-county descriptive report records a 22-county pilot, beyond the acquisition-only account in the earlier map. It distinguishes applicant-weighted and equal-county summaries, records provisional name/score linkage, excludes outside-county placements from the outcome denominator while retaining them as peers, and retains confirmed unassigned applicants in the denominator. Its scope is selected admission-round applicants, not all eighth graders or a national population.

This update reads the saved report; it does not independently re-audit its county sources or establish a model fit. The report explicitly distinguishes observed success fractions from institutional K/N and descriptive associations from causal effects. LASZLO’s domain work continues separately. No new county work or analysis is launched here.

6. The next scientific and writing sequence

Agreed working method: One question at a time, with a written record of the baseline, change, reason, result and route back. Preserve negative or inconclusive findings. The aim is to learn which explanation the evidence supports, not to tune bands until an expected curve appears.

The following is our proposed implementation of Alex’s direction. It is a plan, not a claim that the analyses have been done or permission to start a broad sweep.

1. **Finish interpreting the current BPM comparison.** Start with the top-band shared-versus-LOO pair and its counts. Record what it tells us about the proposed congestion-sensitive band and what remains confounded by performance range, minutes, selection and missingness. Keep the 2.5% alternative, new minutes cutoffs and further significance tests separate until chosen.
2. **Specify the scientific curve and its success criteria.** Decide the horizontal environment measure, the outcome probability, and whether own performance is fixed, restricted to a band, or averaged over a declared population. Use one coherent definition across domains where the data permit, recording genuine domain differences. For the proposed peak/curvature summaries, specify units and whether curvature means a local second derivative or a quadratic approximation. If no interior peak exists, report that rather than force a maximum or negative squared coefficient.
3. **Carry the mathematics through selection.** Begin from the score and Gibbs weight, but retain the changing normalizer and all other affected scores when performance changes. Distinguish a conditional next draw from final inclusion after K draws. Fix the other inputs explicitly; a team mean or pressure alone does not determine an entire selection curve without assumptions about the rest of the population. Check monotonicity and any needed parameter restrictions rather than treating Alex’s expectation as a proof. Preserve the difference between the teammate near theta and the focal player near the selection boundary.
4. **Establish why each added ingredient is needed.** First inventory the existing SCOUT, COMPASS and model experiments against the stated claim. Then use analytic comparisons or the smallest necessary controlled experiment to show what is lost when an ingredient is removed. A changing winner set is not a downturn criterion. Nor does a monotone individual rule settle an aggregated environmental curve without assumptions about sorting and composition.
5. **Show parameter effects in comparable curve families.** Vary a declared parameter while stating what remains fixed, including the threshold rule when scarcity changes. Put directly compared curves on the same axes and report support, uncertainty and any peak/curvature changes. Keep mosaics as diagnostics where useful. New experiments need their own question and record; the current empirical mosaics do not complete this model-based task.
6. **Revise the manuscript around that evidence.** Resume the paused 4.4.3 paragraph and model summary, settle terminology without unapproved renaming, then develop 4.5 and 4.7 around the target curves and necessary ingredients. Replace the rejected figure presentation with selected, verified comparisons. Reconcile the opening roadmap with the actual order. Complete the whole-section style pass after content is settled; organize the appendices and field numbering separately.
7. **Specify fitting and a testable prediction.** Write the likelihood for the chosen observation and selection process, resolve the historical fitting discrepancies, and state a prediction before examining the comparison meant to test it. Threshold/band location and system noise are the candidate families from PD48. An exploratory match after inspecting several plots is not yet an independent prediction test. No completed exact-K refit or cross-domain predictive validation is established by this map.

Presentation and scheduling

For the defense, retain the useful player imagery but shorten the machinery and make room for why the model is needed, what fitting reveals, and what predictions survive testing. This is the post-brief presentation direction, not an instruction to strip necessary derivations out of the written dissertation. Administrative dates, committee arrangements and any extension remain to be confirmed by Charles; PD47 discussed possibilities rather than approving a schedule.

7. Source register and revision record

Direct advisor evidence read for this update: PD47 (October 16), PD48 (October 20), PD49 (October 23), and the post-brief feedback transcript, linked in Section 1 with relevant timestamps. The PD44/45 account is retained as historical context; later explicit guidance supersedes it where identified. Transcripts are fallible records, not verbatim-certified institutional decisions.

Current manuscript evidence: The saved Word working draft and retained comments, inspected October 26 with deleted text excluded and mathematical text included. Section 3 records present content and remaining requests separately. An advisor's comment is evidence of a request, not proof that the associated edit is complete. This update does not claim a line-by-line attribution of each revision to a specific meeting.

Scientific and empirical records: The October 18 LOO/HERO handoff; the current NCAA walkthrough through Entry 7; PPM and BPM notebook settings and run pointers; aligned BPM run `20261025_191833_184458`; the October 25 Romania selected-county report; and the LG decision log. Saved outputs were read, not recomputed. The historical likelihood and mechanism evidence trails remain the model plot trail and the MLE/Bernoulli response; their existence does not constitute a new validation of all old results.

Revision record: The September 30 map was expanded on October 13, and its NCAA comparison note was subsequently updated through October 23 without changing the top-level date. This October 26 revision updates the advisor direction, later placement of survival analysis, actual manuscript status, empirical branch history, and outstanding scientific tasks. It preserves the existing filename and backs up the preceding Markdown under `manuscript_backups/work_map_before_PD47_PD49_update_*`. A source fingerprint record is saved beside that backup to identify what was reviewed. The transition memo changed during this review and was reread; its added October 25 handoff is acknowledged above, while its older section descriptions remain historical.

Files not edited by this task: The Word manuscript, comments, tracked changes, equations, captions, notebooks, data, experimental outputs and transition memo. The PDF beside this map has not been regenerated and is not the current planning record. This Markdown is the updated map; the NCAA walkthrough remains the companion experiment journal.